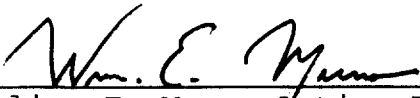


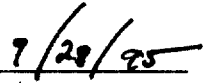
FIVE YEAR REVIEW REPORT
NORTHERNAIRE PLATING COMPANY
CADILLAC, MICHIGAN

Pursuant to CERCLA

PREPARED BY:
U.S. ENVIRONMENTAL PROTECTION AGENCY
REGION 5
CHICAGO, ILLINOIS



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Date

I. INTRODUCTION

A. Authority and Purpose

The U.S. Environmental Protection Agency (U.S. EPA), Region 5, conducted this statutory five-year review under Section 121(c) of the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA). The purpose of a statutory five-year review is to evaluate whether a completed remedial action remains protective of human health and the environment at sites where hazardous waste remains on-site at levels that do not allow for unlimited use and unrestricted exposure. The Type Ia review conducted for this site is applicable to a site at which response is ongoing. This review will be placed in the Site files and local repository for the Northernair Plating Company Superfund Site (the Site) in Cadillac, Michigan.

B. Site History

The Site is located at 1002 Sixth Street in the Cadillac Industrial Park, an area approximately 1 mile square within the City of Cadillac. It is bordered by industrial, municipal and undeveloped property. The Site was listed on the National Priorities List (NPL) in July of 1982. In 1983, a removal action was conducted to remove drums and a sewer line from the site. Subsequently, a Remedial Investigation/Feasibility Study (RI/FS) was conducted by the MDNR, which determined that remedial actions were needed to protect public health and the environment from contact with contaminated soils (Operable Unit 1). A Record of Decision (ROD) was signed for Operable Unit 1 of the Site on September 11, 1985.

An additional RI/FS was conducted on the area-wide groundwater contamination, which determined that remedial actions were also necessary for contaminated groundwater (Operable Unit 2). The RI/FS determined that the groundwater contamination associated with the Northernair Site was co-mingled with that associated with the Kysor Industrial Corporation Superfund Site. A ROD was signed for Operable Unit 2 of the Northernair Site on September 29, 1989. This ROD required the groundwater portion of the Northernair Site to be addressed in tandem with the groundwater contamination associated with the Kysor Site.

Remedial Design and Remedial Action for Operable Unit 1 of the Northernair Site were performed by an MDNR contractor under a Cooperative Agreement with U.S. EPA.

Remedial Design for Operable Unit 2 of the Northernair Site and the entire Kysor Site was performed by a PRP in response to a unilateral Administrative Order (UAO) issued May 16, 1990. Remedial Action for Operable Unit 2 of the Northernair Site and the entire Kysor Site is being performed by PRPs in response to

two additional UAOs issued January 30, 1995. A third UAO was issued April 11, 1995 to two additional PRPs, who are not complying with the order. A Local Development Finance Authority associated with the City of Cadillac (not a PRP) is financing construction.

II. DISCUSSION

A. Remedial Objectives

The remedial action goals of the ROD for Operable Unit 1 were to minimize risks to public health and the environment from direct contact with contaminated materials and to minimize the migration of contaminants to groundwater. The remedy selected to meet these objectives included:

- * Excavate the soils, sewer line, and sediments contaminated with cadmium and chromium to response objectives of 50 mg/kg chromium and 10 mg/kg cadmium, and transport to a privately owned, off-site RCRA subtitle C disposal facility;
- * Break up a 30 foot by 10 foot area of the concrete floor and the drywell in the building, sampling the soil and if contaminated, excavate and dispose as above, backfill the excavated area; and
- * Clean the floor of the building of dust and residue which may contain hazardous substances;

The remedial action goal of the ROD for Operable Unit 2 was to address all remaining potential risks to human health and impacts to the environment. The remedy selected to meet this objective included:

- * Install groundwater extraction and treatment system to remove groundwater contamination from the area surround the site;
- * Conduct groundwater monitoring to assess quality of area groundwater; and
- * Impose access and use restrictions.

B. Remedial Action

On-site work on Operable Unit 1 (Source Control) began October 4, 1988 and was completed November 17, 1988. Approximately 1016 cubic yards of soils were excavated from the site, including areas to the west of the Northernnaire building and inside the building. The process sewer line inside the building and private sewer line serving the building were excavated. Collection and analyses of soil samples conformed to the approved Quality

Assurance Project Plan and confirmed that the cleanup objectives were met. The building floor was adequately cleaned and back-filled. Off-site disposal of excavated concrete was delayed until 1991 due to the difficulty of complying with RCRA Land Disposal Restrictions and associated new treatment requirements. The pre-final inspection was held on January 11, 1991, and the final inspection was held on May 8, 1991.

On-site construction on Operable Unit 2 for groundwater cleanup began June 29, 1995 and is scheduled to be completed in April, 1996.

B. ARARs Review for Operable Unit 1

The Focussed Feasibility Study for Operable Unit 1, dated July 1985, recommended cleanup standards based on the proposed Federal limit for land disposal of cadmium (10 mg/kg) and the conservative end of the suggested range for land application of sludges containing chromium (50 mg/kg) from draft guidelines for the Michigan Municipal Sludge Management Program. These recommendations served as the basis for soil clean-up standards which were selected in the ROD.

The remedy as performed for Operable Unit 1 is compliant with the performance standards selected in the ROD. After the ROD was signed, new State soil cleanup criteria were developed pursuant to Michigan Act 307 and again more recently pursuant to Michigan Act 451, Part 201. The Act 451 cleanup criteria are listed in the State of Michigan's Environmental Response Division (ERD) Operational Memorandum #14, Revision 2, and Interim ERD Operational Memorandum #8, Revision 4, which list soil cleanup criteria for various land uses.

The cleanup standards in the ROD are below the direct contact value for chromium and cadmium in the current Michigan criteria for any land use. Therefore, the existing cleanup of the Northernnaire soils is adequate to protect future users of the Site from direct contact with any chromium or cadmium which may remain in the soil.

The cleanup standards in the ROD for Operable Unit 1 also remain adequate to protect groundwater. The surficial aquifer beneath the Site is not used as a source of groundwater on-site. All affected off-site areas have been supplied with city water and a City ordinance is planned to prohibit additional drinking water wells from the area of groundwater contamination. A groundwater treatment system is currently under construction.

The Focussed Feasibility Study reported the results of groundwater modeling using the leaching potential of the metals left in soils at the Site following soil cleanup. Worst case modeling results demonstrated that any remaining metals leaching

to groundwater would reach a concentration equal to the 1985 federal Maximum Contaminant Level (MCL) for chromium (0.05 mg/l) and cadmium (0.01 mg/l) within 30 feet of the facility. Since that time, the MCL for chromium has been raised to 0.1 mg/l and for cadmium has been lowered to 0.005 mg/l. Therefore, groundwater leaching through remaining contaminants in the Northernnaire soils is expected to meet the current MCL for chromium in less than 30 feet from the facility and for cadmium in somewhat more than 30 feet from the facility. Actual monitoring conducted in Spring of 1995 determined that chromium was not detected in the closest (500 feet) near-surface aquifer monitoring well downgradient of the facility. Cadmium was not sampled. Additional sampling of chromium and cadmium close to the Northernnaire facility is recommended below.

An ARARs review for Operable Unit 2 is not included in this five-year review because construction is on-going. An ARARs review for Operable Unit 2 will be performed during the next five-year review.

III. Recommendations

I recommend that shallow groundwater immediately downgradient of the Northernnaire facility be sampled for chromium and cadmium, to confirm the modeling results predicted by the Focussed Feasibility Study. This work might most conveniently be done by PRPs at the same time as shallow sampling at the associated Ingraham property, scheduled for this fall. No additional actions at Operable Unit 1 are necessary at this time.

I recommend continued implementation of remedial action for Operable Unit 2, including completion of construction of the groundwater extraction and treatment system, groundwater monitoring, and implementation of a local ordinance restricting groundwater use.

IV. Statement on Protectiveness

I certify that the remedies selected for this site remain protective of human health and the environment.

V. Next Five-Year Review

The next five-year review will be conducted by October 4, 1998, which is less than five years from the date of this review because this review was conducted more than 5 years from the date of on-site construction mobilization at the Site (October 4, 1988).